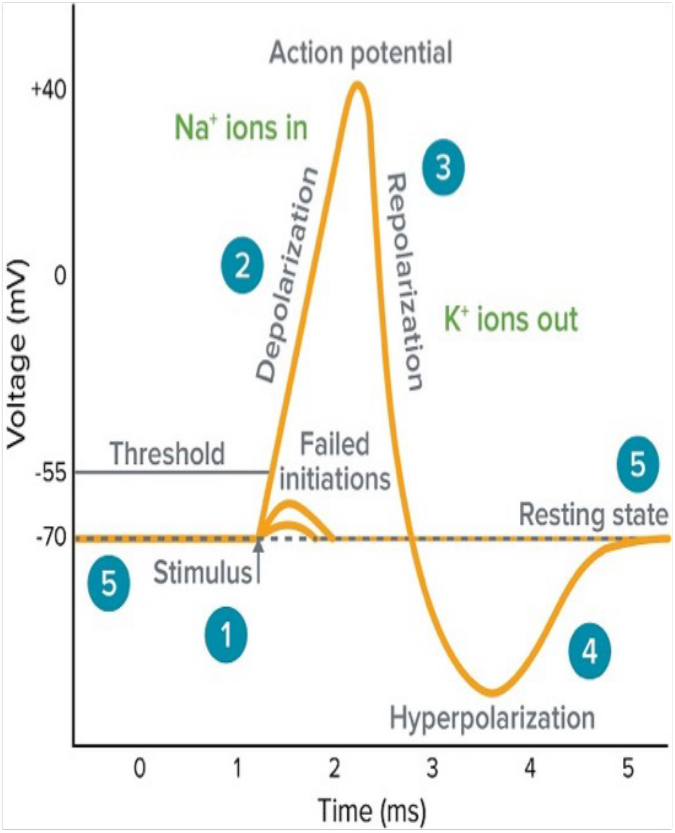
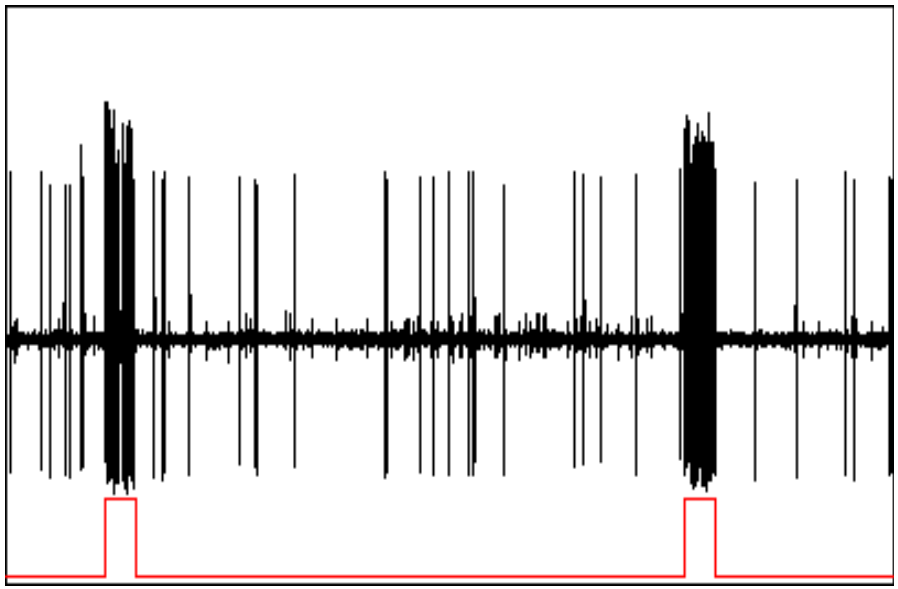
L3

Action Potential



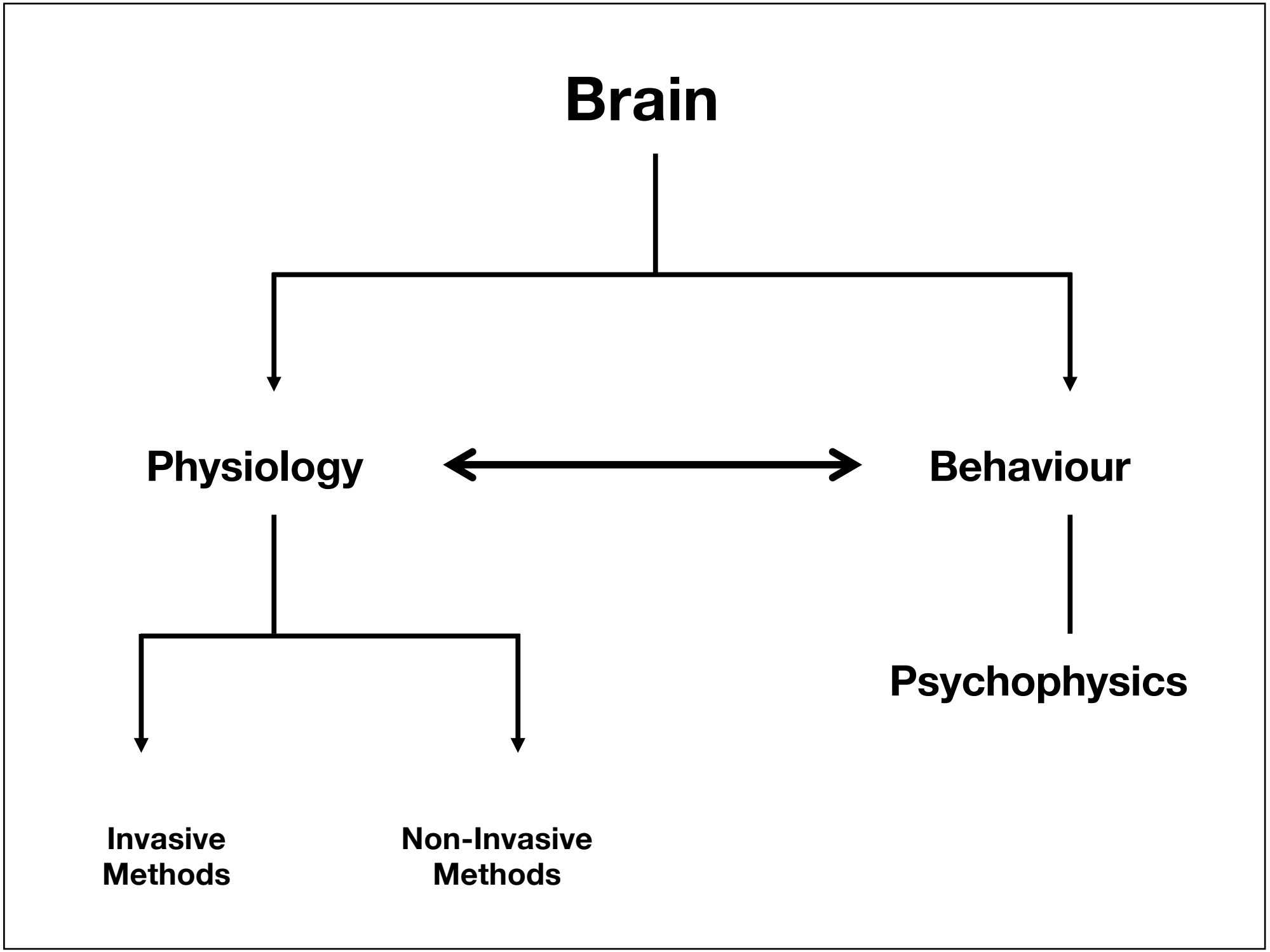
https://www.moleculardevices.com/applications/patch-clamp-electrophysiology/what-action-potential#gref



https://www.ece.rice.edu/~dhj/spike.html

Language of brain

How can we measure Action potentials?


Brain
Physiology
Behaviour
Psychophysics
Invasive Methods
Non-Invasive Methods

The 3 R's of Animal Research

Reduce

Reduce the number of animals used

Refine

Refine tests to cause animals less stress

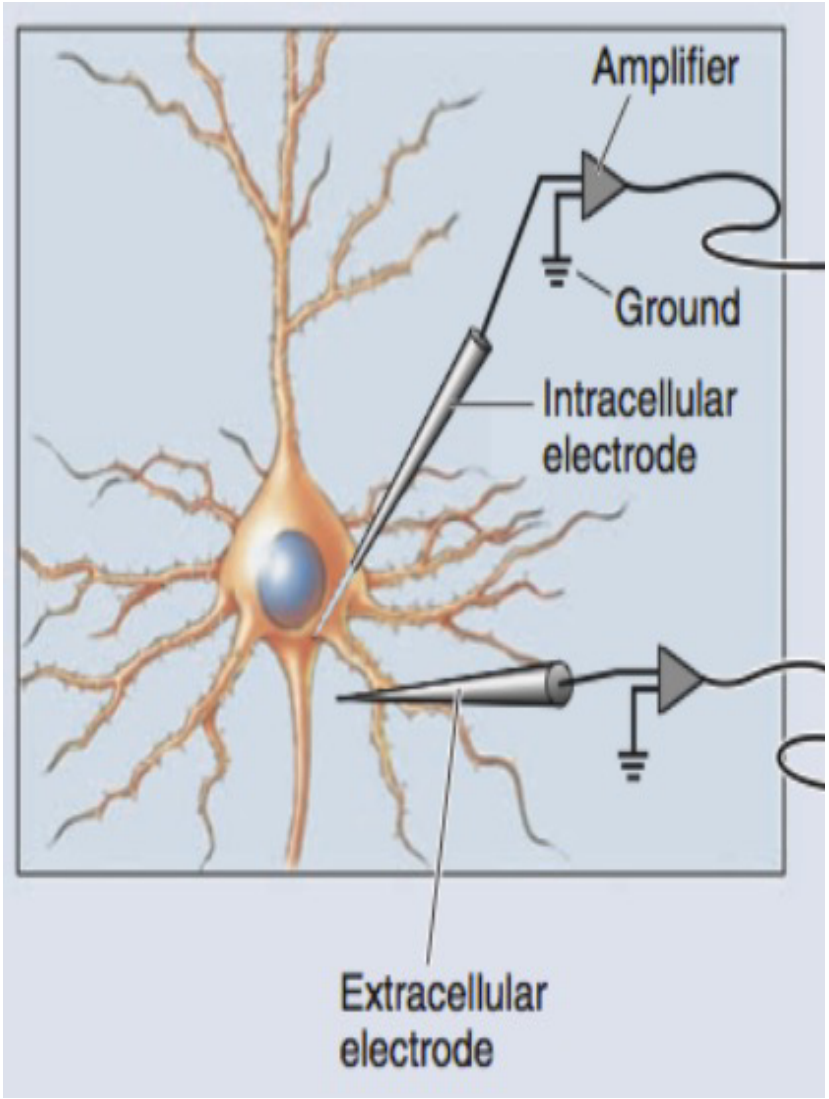
Replace

Replace animal studies with other methods

RockStep
DIGITALIZING IN VIVO RESEARCH

https://www.rockstepsolutions.com/blog/the-3rs-of-humane-animal-research/

Intracellular & Extracellular recording



fMRI(functional Magnetic Resonance Imaging)

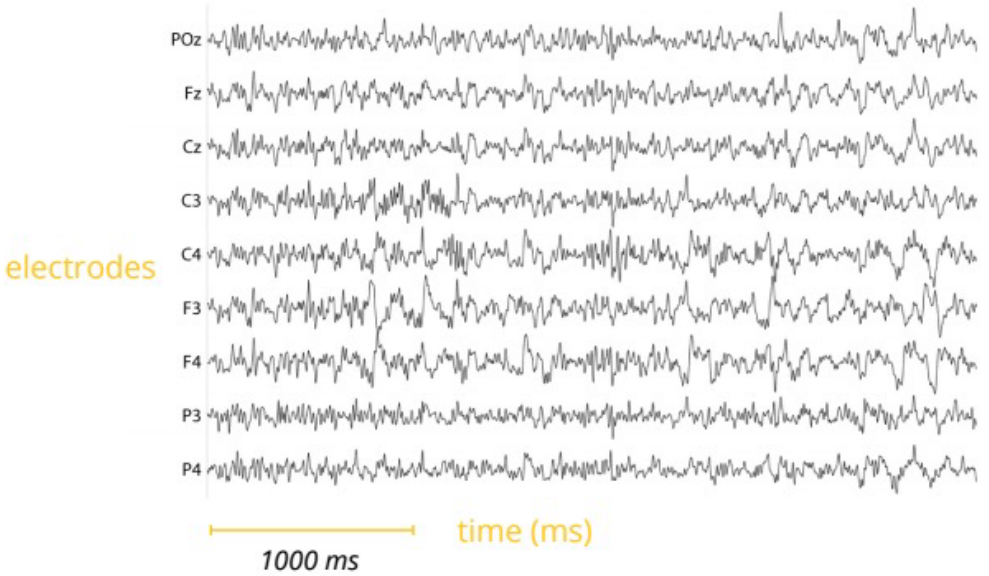
Measures brain activity associated with blood flow.

fMRI(functional Magnetic Resonance Imaging)

- Measures blood oxgenation levels
- Good spatial resolution
- Bad temporal resolution

EEG (Electroencephalography)

https://nld.tamu.edu/eeg



https://imotions.com/blog/eeg/

Non-invasive electrophysiological method to record brain activity.

EEG reflects summed post-synaptic activities of large number of cells.

EEG (Electroencephalography)

- Measures voltage fluctuations resulting from ionic current within the neurons of the brain.
- Bad spatial resolution
- Good temporal resolution

Eye-Tracker

Eye Movements

Pupillometry

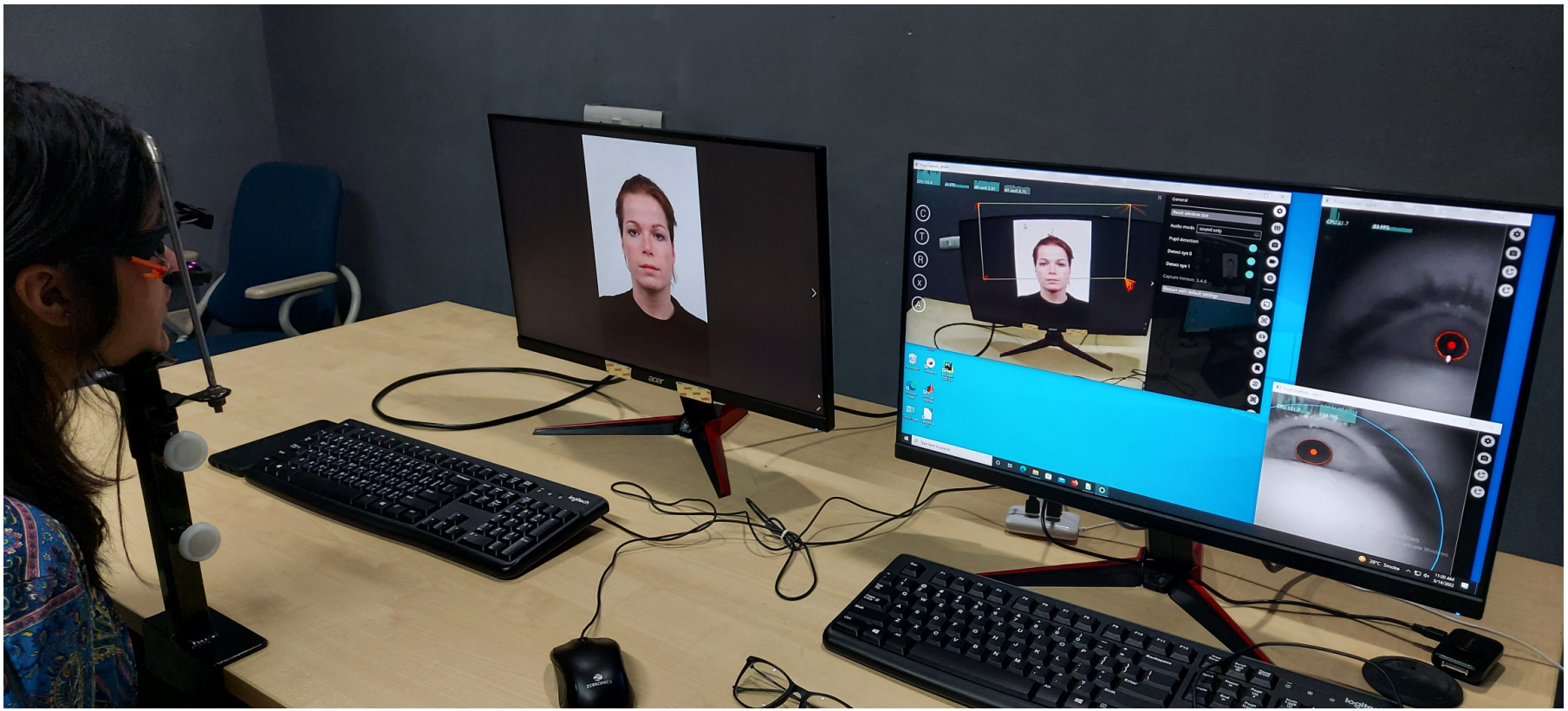
Gaze Maps

Pupil Dynamics

https://www.visioninitiative.org.au/common-eye-conditions/how-your-eyes-work

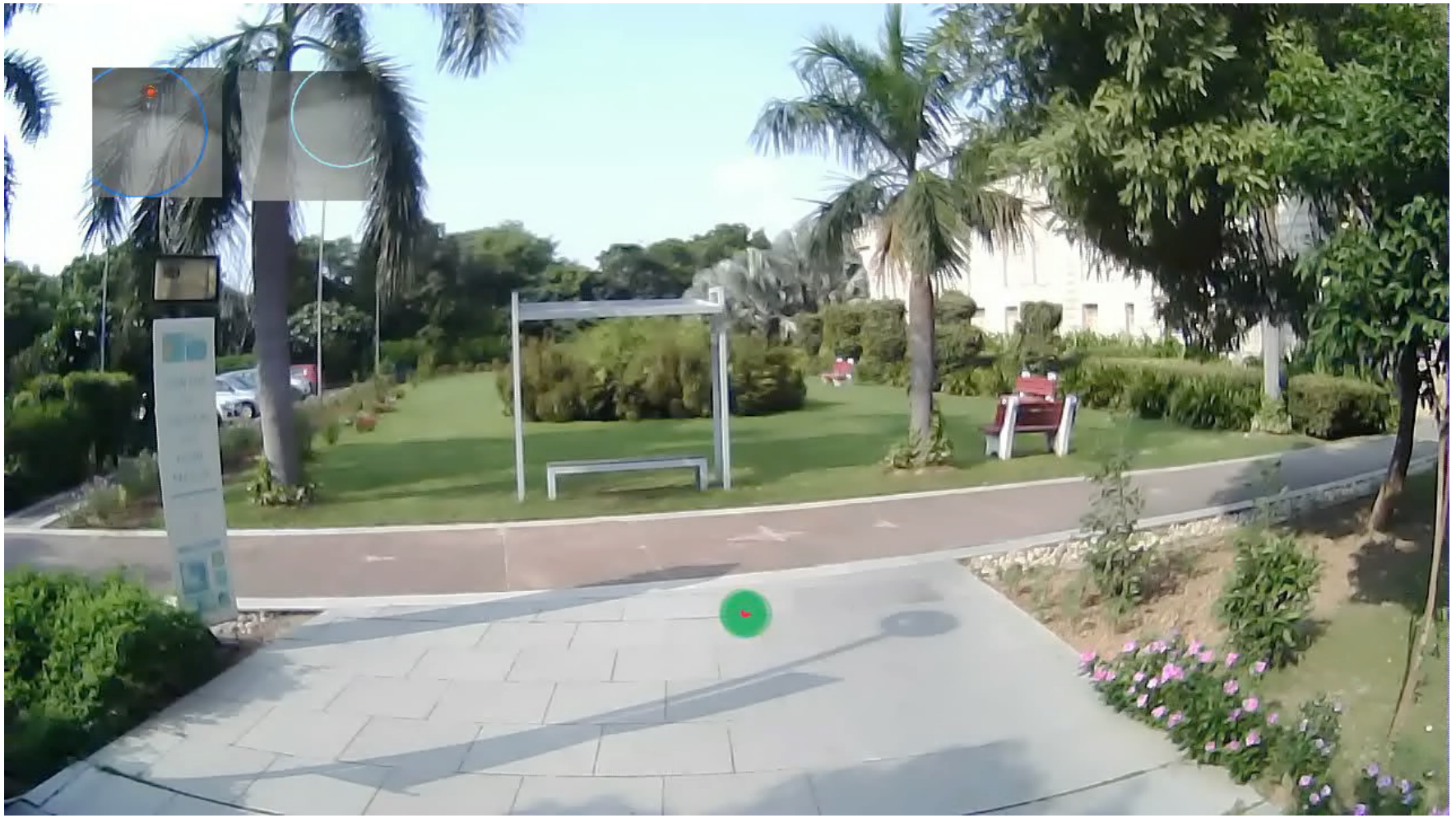
Eye-Tracker Session

Eye-Tracker Session

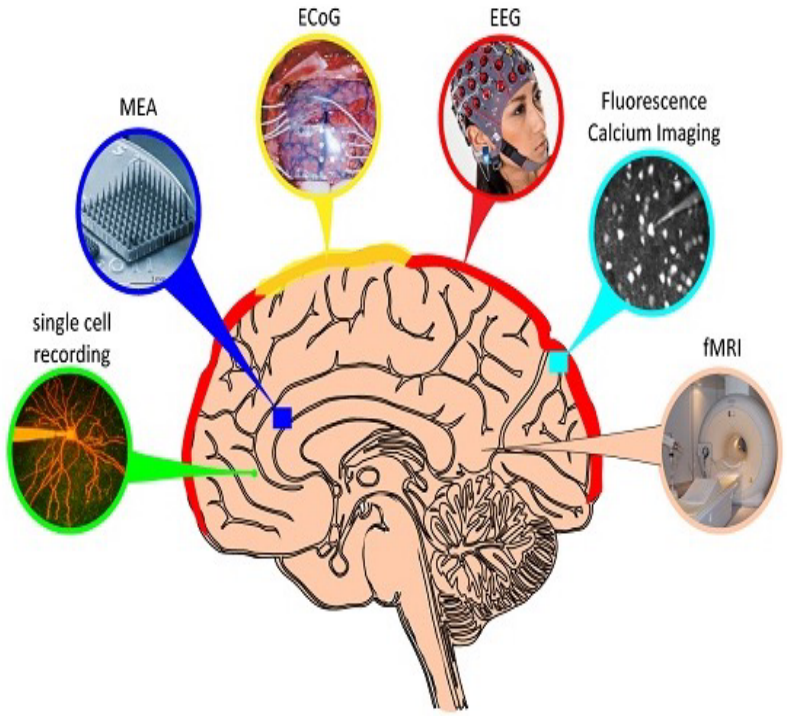


Eye-Tracker Session

Head Movement

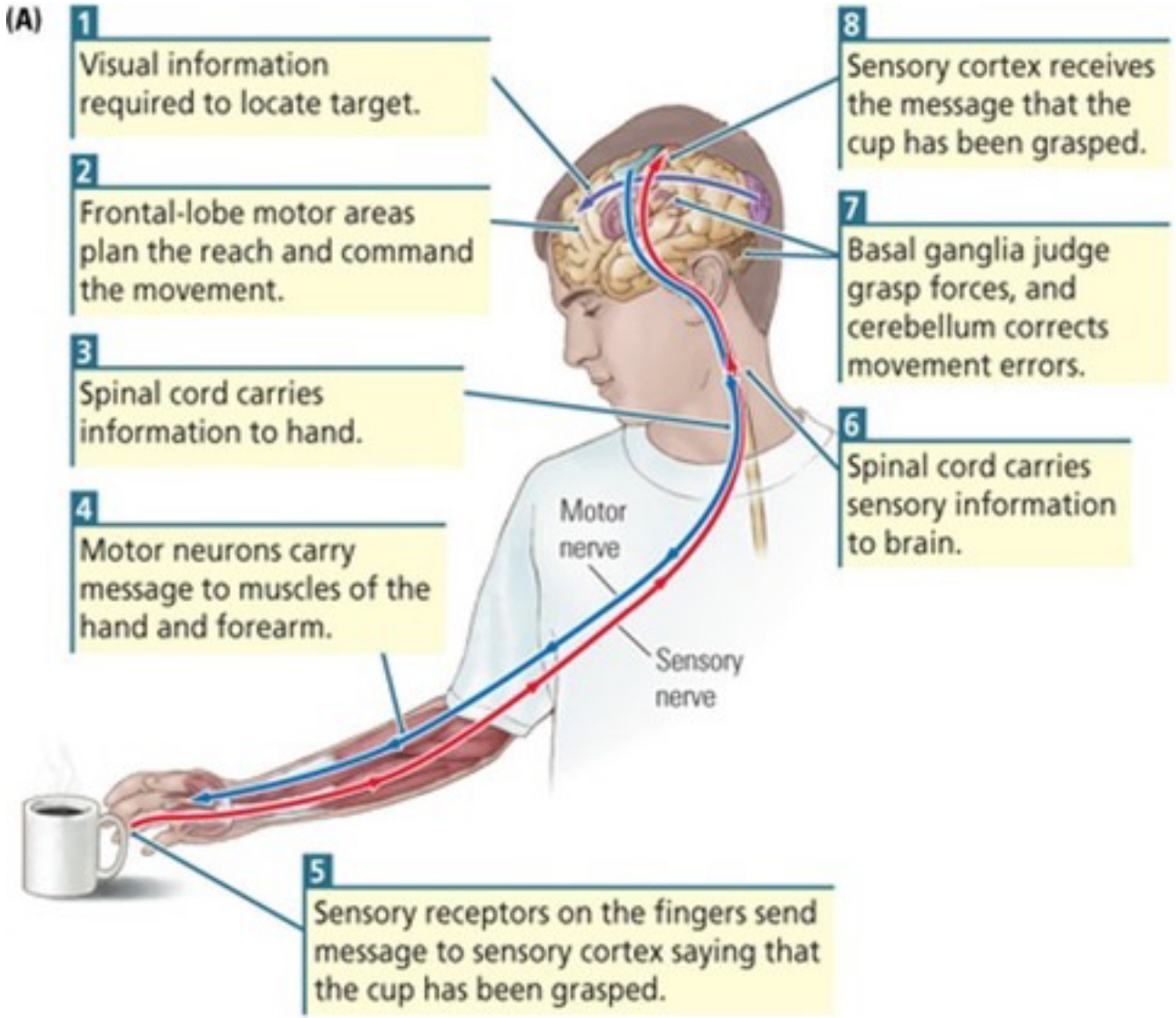


Methodologies to measure brain signals



https://qbi.uq.edu.au/brain/brain-functions/how-measure-brain-activity-people

The motor pathway



https://quizlet.com/234442886/organization-of-the-motor-system-flash-cards/

Motor Movement

1. The *degrees of freedom:* How are movements selected out of infinite options to achieve particular tasks?

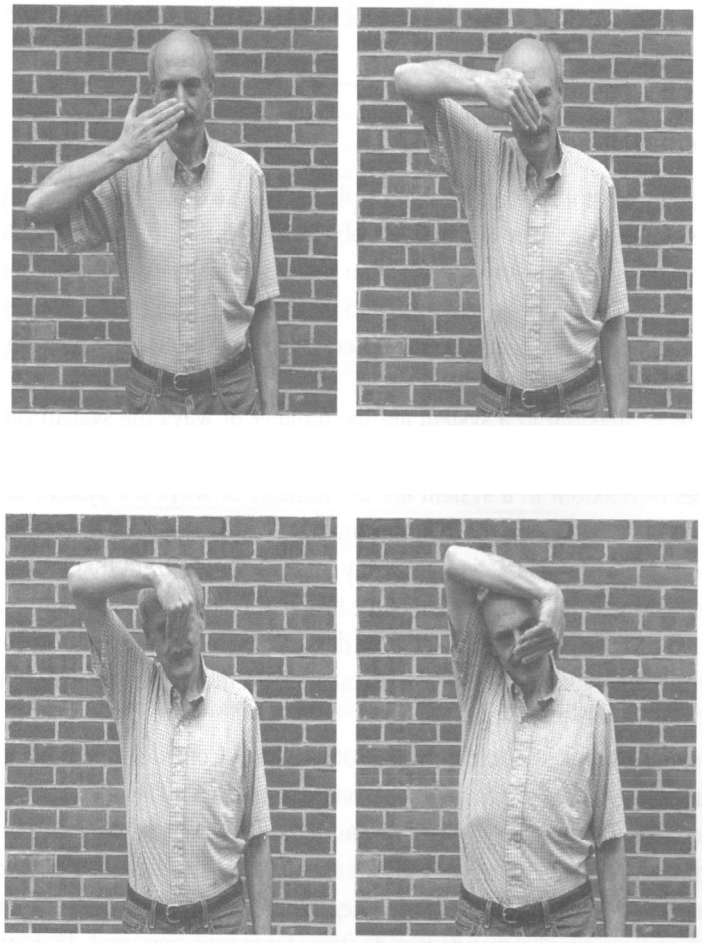
2. The *sequencing and timing :* How are behaviors ordered in time?

3. The *perceptual-motor integration :* How are perception and motor control combined?

4. The *learning :* How are perceptual-motor skills acquired?

THE DEGREES OF FREEDOM PROBLEM

The degrees of freedom in a system are <u>the ways the system can independently vary</u>.



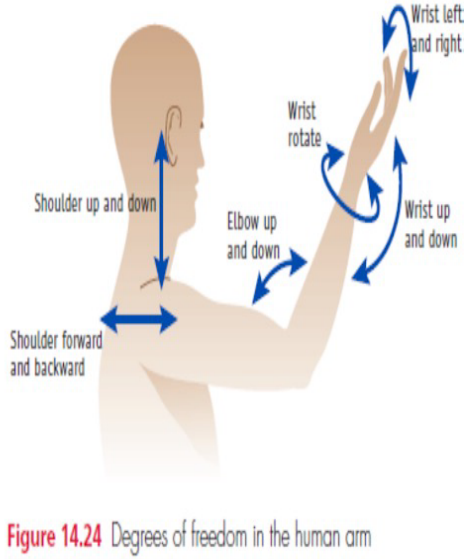
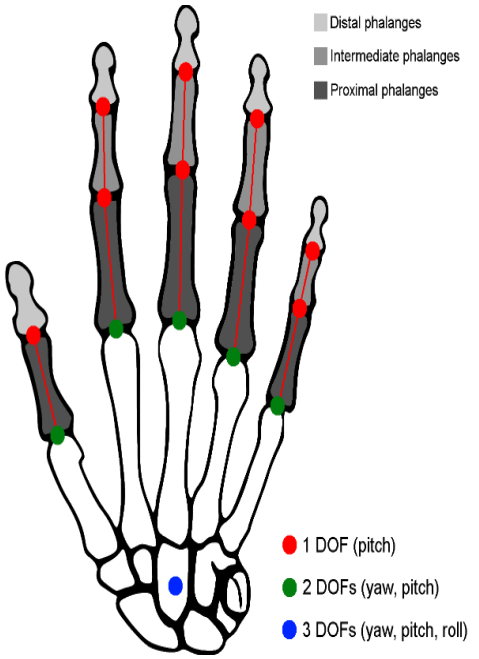


Figure 14.24 Degrees of freedom in the human arm

http://web1.muirfield-h.schools.nsw.edu.au/mahara/view/view.php?id=14089



Moreira et al, 2015

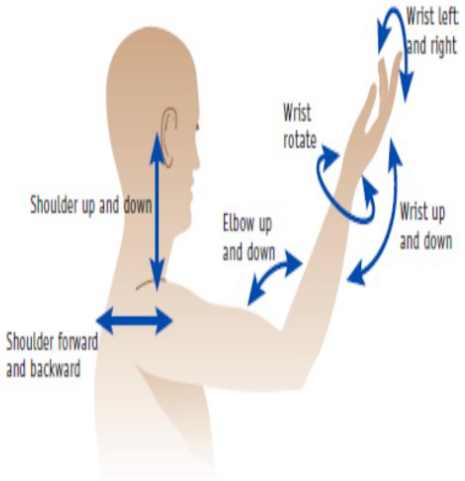
- ***kinematics*** **(positions in time)**
- ***Kinetics*** **(forces in time)**
- **Mapping of neural activity to muscle activity.**

Many-to-one mapping

THE DEGREES OF FREEDOM (DOF) PROBLEM

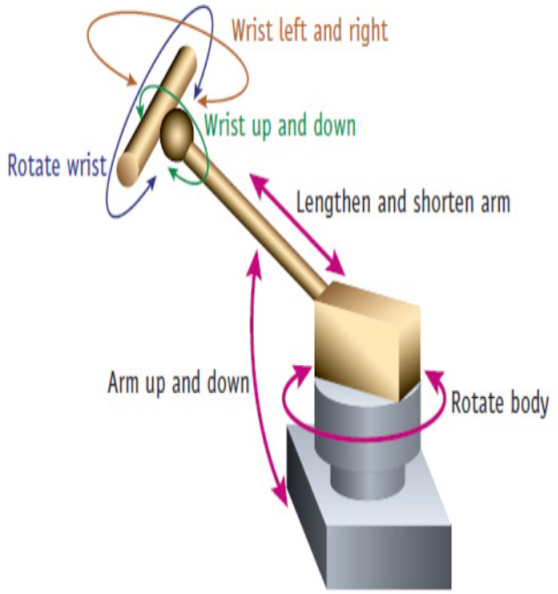
Many-to-one mapping

Degrees of freedom: Human Arm



http://web1.muirfield-h.schools.nsw.edu.au/mahara/view/view.php?id=1408

Degrees of freedom: Robotic Arm



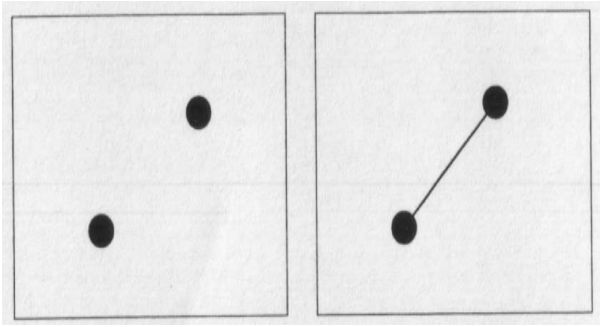
Synergy

~~Cognition~~

Synergies

Interaction/Cooperation

4 DOF



3 DOF

??

Synergies can reduce the number of degrees of freedom that must be independently controlled.

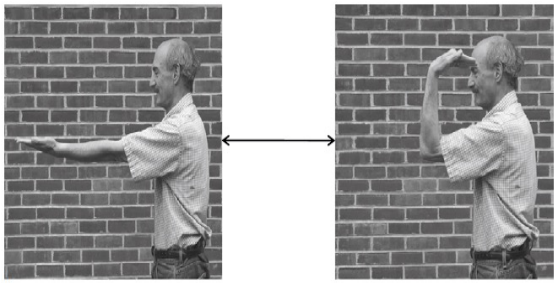
Synergies

https://gfycat.com/equalbeneficialallosaurus

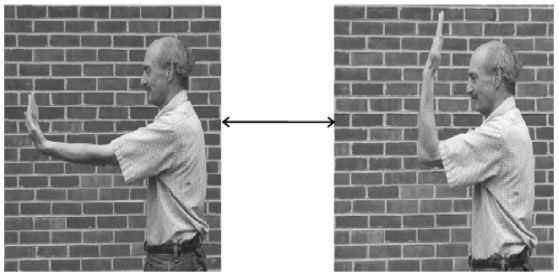
Coordination *within* limbs

Behaviour of one joint influences other

Functional linkages between different parts of the body

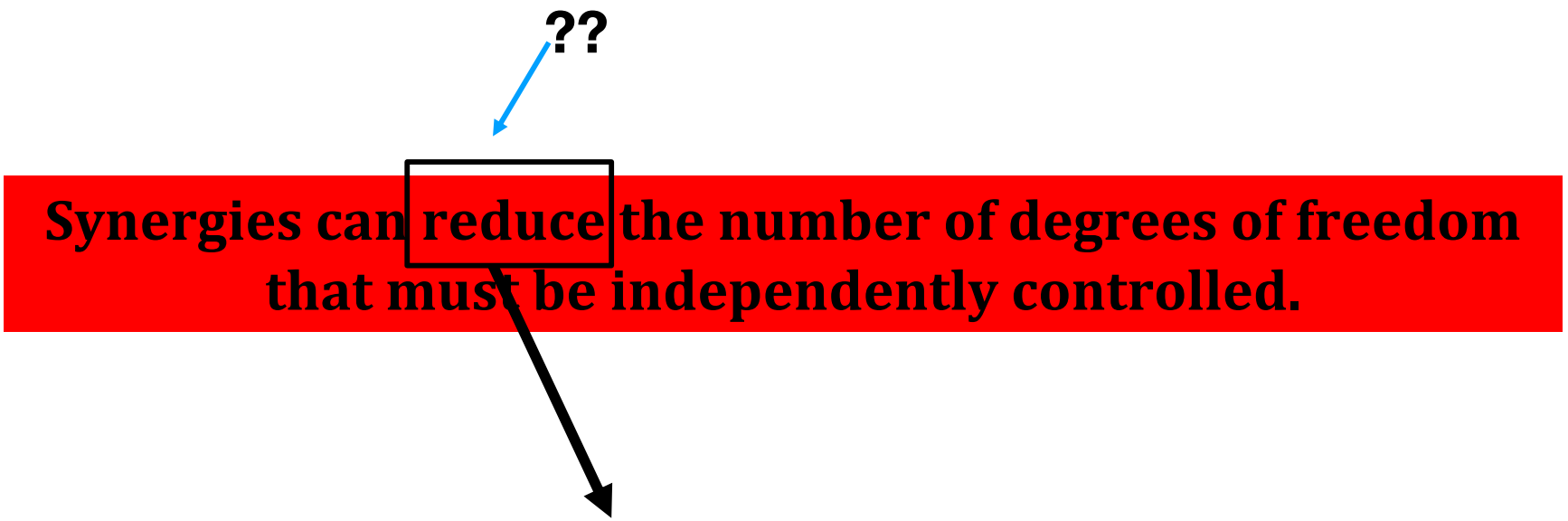


A



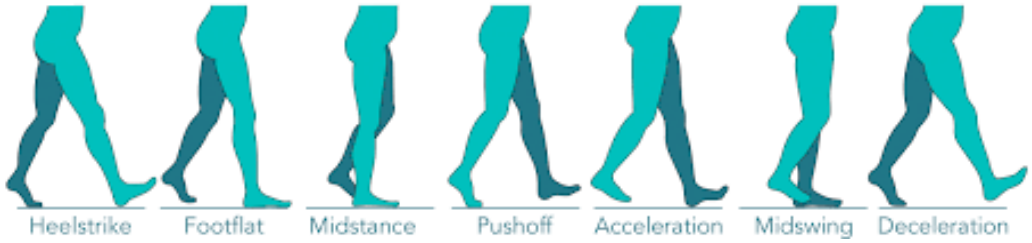
Synergy ⟷ **DOF**

Synergies


??
Synergies can reduce the number of degrees of freedom that must be independently controlled.
Bias neuro-motor control

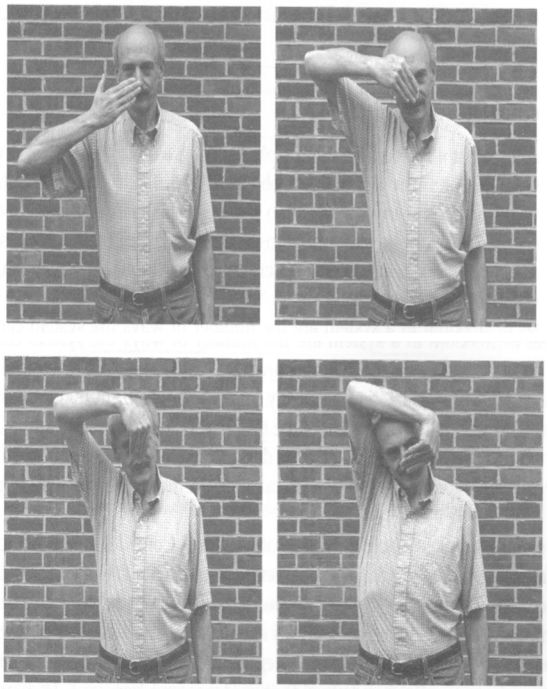
Mechanics & Efficiency

Mechanics



https://www.tekscan.com/blog/medical/gait-cycle-phases-parameters-evaluate-technology

Efficiency



Relying on physical mechanics eliminates need to generate just the right command @ right time

Only more efficient movements are executed

Passive dynamic walker

https://www.youtube.com/watch?v=WOPED7I5Lac

https://www.youtube.com/watch?v=rhu2xNIpgDE&t=15s

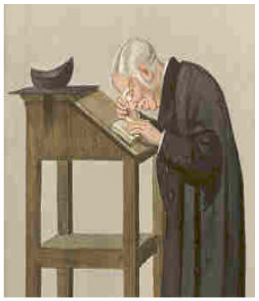
Degree Of Freedom

- Synergy
- Mechanics
- Efficiency

THE SEQUENCING AND TIMING PROBLEM

Control of sequencing and timing of our behaviour

Spooner and Student



William A. Spooner

”roaring pain" instead of ”pouring rain."

A spoonerism is an error in speech in which corresponding consonants, vowels, or morphemes are switched between two words in a phrase.

Action Slips: wrong motor activity

When we engage in behaviors that have distinct elements, such as speaking, typing, or walking, the elements of the behaviors must be ordered correctly in time . How is such ordering achieved?

SEQUENCING: Coarticulation & Timing

Simultaneous motions that occurs in sequential tasks

https://www.online-tech-tips.com/computer-tips/best-methods-to-improve-your-typing-speed/

https://mindfulpresenter.com/7-takeaways-barack-obamas-better-presenter/

https://www.musicradar.com/how-to/improve-your-acoustic-folk-guitar-playing-in-20-minutes

Effective way of increasing response speed